Structured Problem Solving (SPS)

Process Modeling Overview

Process Modeling
Overview

- What is Process Modeling?
- Hierarchy and Modularity
- Constructs
- Essential Model
  - Hierarchy Chart
- Implementation Model
  - Flowchart
  - Pseudocode

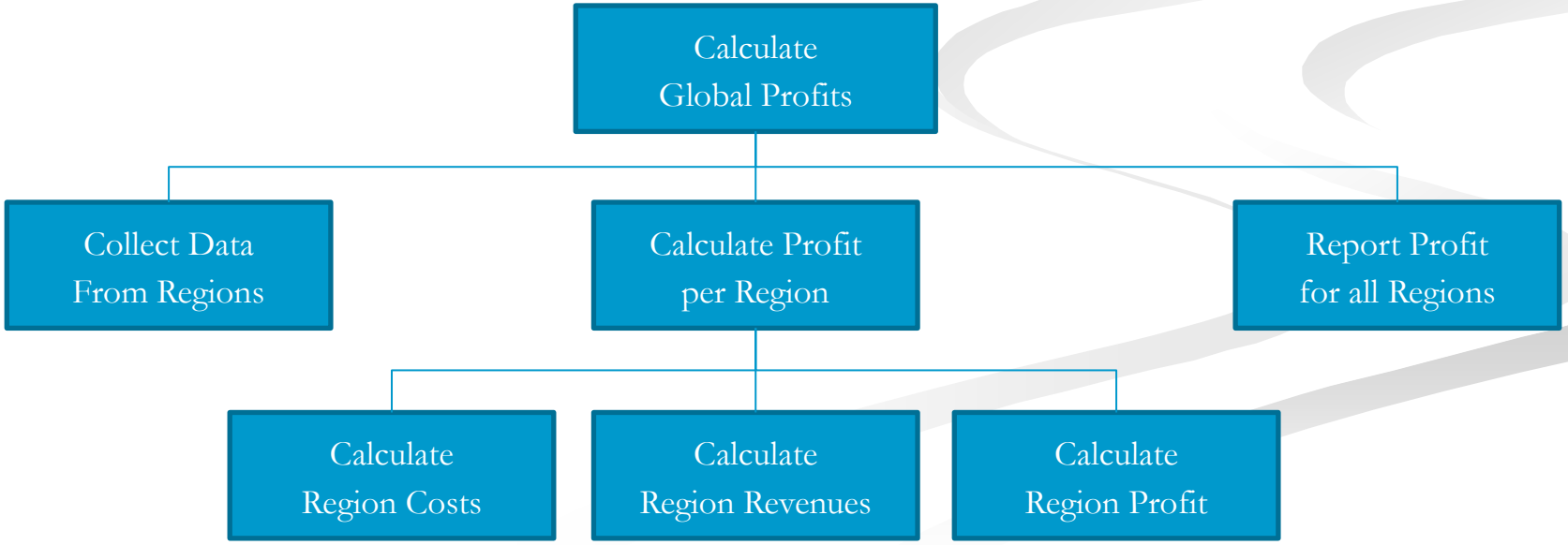
Process Modeling in Structured Problem Solving

What is process modeling?

- Structured Problem Solving (SPS) strategy:
  - Implements decomposition via hierarchy and modularity
- There are processes used to design how to solve problems
- Just like we decompose the problem, the process of solving the problem is itself decomposed via "separation of concerns"
  - The "what" and the "how"
  - Think of "events" and "instructions"
- There are separate process models for the "what" and the "how"

Structure of Models Used in Problem Solving
Hierarchy and modularity – Not just a process flow

- **Hierarchical structures** decompose tasks or processes to model "belonging to" relationships
  - A "child" activity "belongs" to only one "parent" activity
- This readily enables the decomposition of tasks into **modules**
- These can represent "chunks" of work which are done together



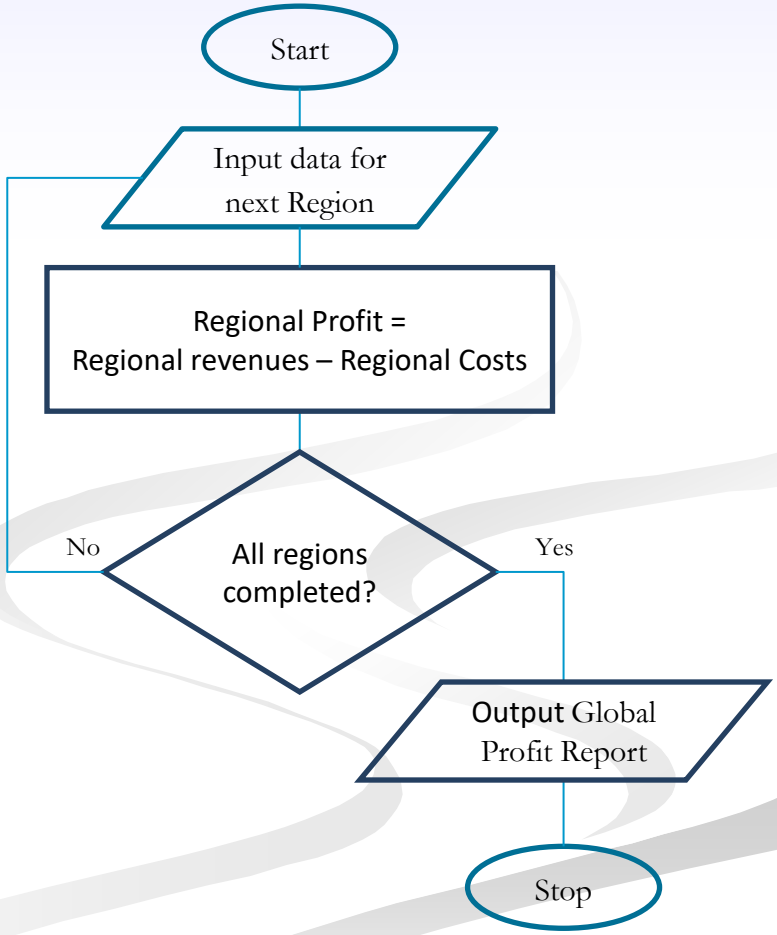
Process Modeling in Structured Problem Solving

What are the "constructs" of programming?

- A program is a sequence of instructions
- A programming language supports structured programming if it supports sequence, selection, and iteration.
- **Sequence**:
  - Ability to execute modules in the sequence necessary to solve a problem.
- **Selection**:
  - Ability to select between executing one module or another depending on a circumstance or condition.
- **Iteration**:
  - The ability to repeat the execution of a module if necessary.

The Use of Models in Problem Solving
How – the Implementation Model for logic

- Flow charts model the logic for actions or steps in solving a problem
- Flow of procedures is dependent on inputs, decisions, and results of prior procedures
- Sequence, selection, and iteration are all displayed in a flow chart.



The Use of Models in Problem Solving
How – the Implementation Model for coding

- Pseudocode is a language agnostic, line by line description of the items and actions in a block of code

- By being language agnostic, the pseudocode is can be used to implement the code in any programming language

- It is also readable by non-programmers to understand how the program will execute

```
main()
{
    lastRegion  = False

    while (! lastRegion) {
        call getRegionData()
        call processRegion()
        update lastRegion
    }

    call outputReport()
    stop
}  # end main
```

```
getRegionData()
{
    input  region data
}  # end getRegionData

processRegion()
{
    calculate region profit
}  # end processRegion

outputReport()
{
    output region report
}  # end outputReport
```